

Problem Definition and Elaboration:

Zoos use enrichment devices to promote natural behaviors, mental stimulation, and physical activity in large mammals¹. However, enrichment tools for large hoofstock such as elands must be safely suspended and positioned to prevent entanglement or injury². Adult elands, such as Loki (male), Olive (female), and Violet (female), can weigh over 1200 pounds and reach approximately 6 feet tall including their horns³. Additionally, common elands are fast runners with a nomadic lifestyle. Despite their bulky, cow-like build, they possess surprising stamina. Over the course of a day, a herd of elands can travel up to 35 miles (55 km) to find new browsing sites. Thus, they require constant activity in captivity, usually with the use of toys. With their toys, they push, lift, swing, and spar using their horns. Indeed, in the wild, young male elands spar by tangling their horns and pushing each other until one is victorious⁴. Due to this behaviour, the elands' horns often catch in the chain link fence of their enclosure, which rips off their antlers⁵.

The Nashville Zoo does not have the current ability to promote their elands' full wellbeing with ease, and has thus requested two enrichment toy hangers to be installed outside of the elands' behind-the-scenes enclosure. The structure must extend over the currently existing 8-foot fence of the enclosure and safely support toys weighing up to 50 pounds with width up to 1.5 feet, and the toys must sit 1-2 feet inside the enclosure because if the enrichment toys were attached too close to the fence, the elands would be at risk of entanglement with the fence, which could result in injury to the animal or damage to the enclosure.

The zoo requires a durable, long-lasting hanger that can withstand outdoor weather conditions and the force generated by a large animal interacting with suspended enrichment. However, the zoo has informed us that the device need not sustain the weight of the eland if it were to tangle in the toy, as that is a rare occurrence, and if it were to occur, the toy would simply rip. Additionally, since the hanger will not be on public exhibit, aesthetic concerns are secondary to safety, stability, and structural integrity. The device must be installed outside the enclosure and may be inserted into the ground for additional support.

Although this design centers on the Nashville Zoo's elands, many zoological facilities house large hoofstock species that require safe suspended enrichment. A successful solution would contribute to improved enrichment design practices across similar institutions, supporting animal welfare and responsible habitat engineering.

Problem Identification:

The problem is to design and create two ground-mounted enrichment toy hangers for the zoo that:

- Are inserted into the ground outside the enclosure.
- Extend safely over the pen's 8 foot fence.
- Support enrichment toys weighing up to 50 pounds that are 1.5 feet wide.
- Prevent eland entanglement with fence.
- Withstand outdoor weather exposure and repeated animal interaction.
- Maintain long-term durability with minimal maintenance.
- Contain an eye bolt or carabiner attachment system to suspend enrichment toys safely.
- Remains stable under dynamic loading conditions caused by the elands' playing.
- Can change the height of the toy's center to ~5 feet off the ground and be 1-2 feet inside the enclosure.

Two installation locations are specified: Male enclosure (Loki) and female enclosure (Olive and Violet).

Stakeholder Concerns and Connections:

Primary Stakeholders:

- Zoo Animal Care Staff: They benefit from a safe, durable, and low-maintenance toy hanger that allows easy attachment and removal of enrichment devices.
- Elands: They benefit from the structure's ability to enhance their enrichment and to limit their fence entanglement.
- Zoo Veterinarians: They benefit from the reduction of injuries such as horn entrapment in the fence.
- Zoo Maintenance Staff: They benefit from a durable design that minimizes repairs and withstands environmental exposure.

Secondary Stakeholders:

- Zoo Administration: They benefit from a cost-effective, long-lasting solution that supports animal welfare goals.
- Other Zoological Institutions: They benefit from a successful design model for similar enrichment programs at their institutions.

Problem Justification:

Large ungulates require environmental enrichment to maintain psychological well-being and encourage natural behaviors such as pulling, lifting, and foraging. However, suspended enrichment devices introduce mechanical risks when improperly designed⁶. A 1200+ pound animal interacting with a 50-pound suspended object can generate significant dynamic forces, increasing the need for structural stability and secure anchoring⁷. Additionally, elands have long, curved horns and considerable height (approximately 6 feet), increasing the risk of entanglement if chains or attachment hardware are improperly positioned. The hanger must therefore be elevated appropriately above the 8-foot wall and designed to eliminate loops, sharp edges, or gaps where horns or limbs could become caught. Installing the structure outside the enclosure reduces direct animal interference with the base support but requires sufficient extension and cantilever strength to prevent bending or tipping. Ground insertion adds stability but requires careful material selection to prevent corrosion and structural degradation over time. Designing a safe, durable enrichment hanger directly supports animal welfare, reduces injury risk, and ensures consistent enrichment availability. A well-engineered solution enhances habitat functionality without requiring major enclosure modifications, making it a practical and scalable improvement for zoological facilities.

Citations

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3. The Zoo Briefing
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